

MotionScorer:

Reading Trajectory Quality out of a Frozen Driving VLM

One exact attention identity → classical constrained filtering → nuGuidance

Focus

Turn one architectural identity into a scoring method: the score query's context-attention branch *is* a tap-weighted filter over frozen values, so eliciting a trustworthy score becomes a *constrained readout-selection* problem, not a finetuning problem. The slides follow the derivation and label every step by how much it is proven.

MotionScorer / nuGuidance draft (ICLR 2027, under review); Capon (1969); Van Trees (2002); Lou et al. (SLQ, 2026).

1. The Missing Neural Critic

A world-model view of driving gives one representation three roles: a **Driver** that plans, a **Simulator** that imagines, and a **Critic** that judges. Two are neural; validation still leans on fixed rule tables.

What a driving VLM already does

- ▶ Reads the scene and explains *why* a maneuver is unreasonable.
- ▶ Its semantic output tracks the true evaluation.

What it does not do

- ▶ Directly prompted numeric scores are unstable and collapse to a few round values.
- ▶ Existing fixes rewrite the evaluator: finetuning, reward heads, rule-gated authority.

The question we actually ask

Not *whether* judgment is present, but *through which channel* it can be read out without touching the weights.

Source: MotionScorer draft, §1 and App. C (motivation).

2. Three Observations That Force the Design

The method is not proposed; it is *cornered* by three measurements (App. C). Each observation removes one design freedom.

(i) Judgment is present	(ii) Prompting collapses	(iii) Target is underdetermined
A linear probe on frozen hidden states separates good/bad trajectories far above the rule score. ⇒ <i>do not add capacity; read what exists.</i>	Directly prompted scores pile onto a few integers. ⇒ <i>the failure is the readout channel, not the representation.</i>	Different readouts fitting the <i>same</i> pseudo-score diverge out-of-distribution. ⇒ <i>fitting the label does not fix the readout.</i>

Consequence

“The question is not whether judgment is present, but which readout to trust.” Observation (iii) is the seed of the paper’s one genuine open problem: *fit underdetermines the readout*.

Source: MotionScorer draft, App. C (three binding observations).

3. The Exact Identity: A Trailing Query Reads, It Does Not Rewrite

Append a score query after the frozen context under the causal mask. Two facts hold *exactly*, by architecture.

Non-interference (bit-identical context)

Under the causal mask the appended query attends to the context but the context never attends back, so every frozen context state is unchanged (App. I.1 reports a bit-identical check).

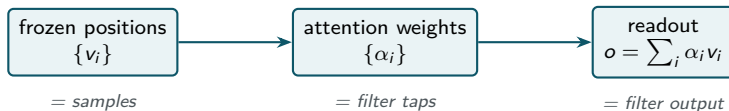
Readout is a weighted aggregation of frozen values (Eq. 2)

$$o = \sum_{i=1}^n \alpha_i v_i, \quad \alpha_i = \frac{\exp(q^\top k_i / \sqrt{d})}{\sum_j \exp(q^\top k_j / \sqrt{d})}, \quad \sum_i \alpha_i = 1, \quad \alpha_i \geq 0.$$

Values v_i and keys k_i are *frozen*; the only handle is the query q , which sets the weights α . This is the whole factual base — everything after it is analogy, assumption, or empirics, and the slides say which.

4. Formula Similarity: This Aggregation *Is* a Filter

Eq. 2 has the exact form of a classical tap-weighted filter — not an approximation of one.



The dictionary is literal

- ▶ positions $\leftrightarrow$ samples
- ▶ attention weights $\leftrightarrow$ taps
- ▶ $\sum_i \alpha_i = 1 \leftrightarrow$ distortionless / unit-gain constraint

One caveat, stated up front

Classical filters *assign* taps freely. Here taps are *generated* by a frozen network from the query — a constraint we must carry through the whole derivation. **[Assumption]** nothing yet; just a structural fact.

Source: MotionScorer draft, §3; Capon (1969); Van Trees (2002), Ch. 6.

5. The Borrowed Criterion: Deflection and Its Closed-Form Optimum

Filtering theory answers “which taps?” with the *deflection* ratio of signal power to interference power.

(classical, imported) Deflection, maximizer, and the two-line proof

$$\rho_{\text{free}}(\alpha) = \frac{(\alpha^\top \theta)^2}{\alpha^\top \Sigma \alpha}, \quad \alpha^* \propto \Sigma^{-1} \theta, \quad \rho_{\text{free}}(\alpha^*) = \theta^\top \Sigma^{-1} \theta.$$

Factor, then Cauchy–Schwarz: $(\alpha^\top \theta)^2 = ((\Sigma^{1/2} \alpha)^\top (\Sigma^{-1/2} \theta))^2 \leq (\alpha^\top \Sigma \alpha)(\theta^\top \Sigma^{-1} \theta)$, so $\rho_{\text{free}}(\alpha) \leq \theta^\top \Sigma^{-1} \theta$, with equality iff $\Sigma^{1/2} \alpha \parallel \Sigma^{-1/2} \theta$, i.e. $\alpha \propto \Sigma^{-1} \theta$ (App. C.1).

Honest ledger

This is the Capon/MVDR optimum — *imported*, not discovered. The depth here must come from what the constraint does to this optimum.

Source: MotionScorer draft, App. C.1; Capon (1969); Van Trees (2002).

6. Why the Classical Optimum Is Unusable — and What It Forces

$\alpha^* \propto \Sigma^{-1}\theta$ is demoted to a *reference*. Three obstructions, each forcing a design choice — the “method forced by the scenario” step.

Obstruction	Forced design choice
θ, Σ live after a nonlinear front end — not directly observable	$\Rightarrow$ Estimate a <i>teacher-conditioned</i> statistic ($\hat{s}_E, \hat{D}_E$) instead
Taps are <i>generated</i> by frozen E , not freely assigned	$\Rightarrow$ Optimize only inside the realizable family $\mathcal{A}_\Phi$
One shared readout must serve the <i>whole</i> distribution	$\Rightarrow$ Optimize a distribution-level objective, not per-instance taps

Two-objective split this induces

What to predict (teacher-conditioned signal) vs. *how to read* (stay realizable) — mapping one-to-one onto observations (i)–(iii).

7. The Optimization Space: Realizable Readouts $\mathcal{A}_\Phi$

The feasible set is not all weight vectors on the simplex; it is only those a frozen network can *produce* from a query bank Φ .

Realizable family (Eq. 5)

$$\mathcal{A}_\Phi = \left\{ \alpha(x) = \text{softmax}(q_\phi^\top K(x)/\sqrt{d}) : \phi \in \Phi \right\} \subseteq \Delta^{n-1}.$$

The taps are a *function of the input* through frozen keys $K(x)$; the only free parameter is the query ϕ .

[Assumption][$\rightarrow$] near-tautology: Proposition 1

“Any readout obtained by optimizing E lies in $\mathcal{A}_\Phi$ ” — true, because $\mathcal{A}_\Phi$ is *defined* as that image. It certifies feasibility but proves nothing structural. Contrast LatentGauge’s T_γ : a pointwise algebraic criterion, an explicit closed-form family, and a provable inert control. **This is where depth is missing.**

Source: MotionScorer draft, §4.2, Eq. 5, Proposition 1.

8. The Load-Bearing Step Is an Assumption, Not a Theorem

The filtering view rests on one decomposition. The paper labels it honestly.

[Assumption] Signal-plus-noise model (Eq. 7) — “sole assumption”

$$v_i = \theta_i + \varepsilon_i, \quad \theta_i = \text{quality-aligned component}, \quad \varepsilon_i = \text{nuisance}, \quad \mathbb{E}[\varepsilon] = 0.$$

The classical substitution $(\theta, \Sigma, \alpha) \rightarrow (\hat{s}_E, \hat{D}_E, w)$ “does not inherit the classical guarantee” (paper’s own words); its benefit is therefore **[Empirical]**, tested, not proven.

Why this is still good practice

The assumption is *isolated and named* in the claim registry with a kill switch, so a reviewer knows exactly which single premise carries the method. LatentGauge puts a *proven* proposition in this slot; MotionScorer puts a *declared, falsifiable* one. Different rigor, comparable honesty.

Source: MotionScorer draft, §4.3, Eq. 7; claim registry (“Model — sole assumption”).

9. The Realizable Objective: Teacher-Conditioned Deflection

Replace unobservable (θ, Σ) with teacher-conditioned estimates and maximize deflection *inside* $\mathcal{A}_\Phi$.

Teacher-conditioned deflection and the training objective

$$\rho_{\text{TCD}} = \hat{\mathbf{s}}_E^\top \hat{\mathbf{D}}_E^{-1} \hat{\mathbf{s}}_E, \quad \min_{\phi \in \Phi} [\mathcal{L}_{\text{score}} - \lambda \rho_{\text{TCD}}].$$

$\hat{\mathbf{s}}_E$: teacher-conditioned signal direction; $\hat{\mathbf{D}}_E = \hat{\Sigma}_E + \varepsilon_D I$: loaded interference estimate.

Free structural insight: the ridge is diagonal loading

The regularizer $\varepsilon_D I$ is not a numerical patch — it is classical *diagonal loading* for mismatch-robust MVDR. Naming it predicts a testable curve: performance is *non-monotone* in ε_D , with the optimal loading moving with teacher-noise level. A cheap “surprise point” the paper does not yet claim.

Source: MotionScorer draft, §4.3; Van Trees (2002) diagonal loading; Reed–Mallett–Brennan (1974).

10. The Estimator: How Algorithm 2 Executes the Optimum

The unobservable (θ, Σ) are replaced by four lines of minibatch moments (batch B doubles as the sample size):

$$\hat{s}_E = \frac{\sum_b \tilde{y}_b \tilde{z}_b}{\sum_b \tilde{y}_b^2 + \varepsilon_s}, \quad r_b = \tilde{z}_b - \hat{s}_E \tilde{y}_b, \quad \hat{D}_E = \text{diag}\left(\frac{1}{B} \sum_b r_b \odot r_b\right) + \varepsilon_D I,$$

where $\tilde{y}_b, \tilde{z}_b$ are centered pseudo-scores and latents. Then $\rho_{\text{TCD}} = \hat{s}_E^\top \hat{D}_E^{-1} \hat{s}_E$, $w^* \propto \hat{D}_E^{-1} \hat{s}_E$ (Eq. 9).

Classical object	Realizable substitute	Guarantee status
signal direction θ	teacher-regressed $\hat{s}_E$	[Assumption] (valid iff Eq. 7 holds)
interference cov. Σ	diagonal residual $\hat{D}_E$	[Assumption] + diag. approx. + loading ε_D
free taps α	latent direction w ; taps stay in $\mathcal{A}_\Phi$	[Exact] (closed form in w ; Rem. 1)

Only the last row keeps the classical property. The substitution “does not inherit the classical guarantee” (paper’s words) — which is exactly why the next three slides are evidence, not theorems.

Source: MotionScorer draft, Algorithm 2, Eqs. 9–10, §4.3.

11. Evidence I: Matched Fit Does Not Control Generalization

Same frozen checkpoint, matched protocol; the only difference is the $\lambda\rho_{\text{TCD}}$ term (Table 3).

The certificate is pinned

- ▶ Pseudo-score RMSE: 0.096 vs. 0.095
- ▶ Pseudo-score SRCC: 0.888 vs. 0.891

By the training criterion the two readouts are *indistinguishable*.

What the consumer sees diverges

- ▶ WOD-E2E pairwise: 66.8% $\rightarrow$ 70.3%
- ▶ Held-out ρ_{TCD} : 6.4 $\rightarrow$ 8.7

Plain readout vs. nuGuidance, out of domain.

Why this is the paper's structural dissociation

Fit (what training pins) and OOD stability (what deployment consumes) are driven by *different coordinates* of the readout — the exact analogue of LatentGauge's "displacement budget does not control metric distortion," currently **[Empirical]** rather than proven. Formalizing it is upgrade A on slide 16.

Source: MotionScorer draft, Table 3, §5.2 (matched-fit protocol).

12. Evidence II: The Attention Pattern Is Causally Responsible

One trained checkpoint, scalar head fixed, *no retraining*; intervene directly on the context-attention branch (Table 5).

Intervention	PairAcc. (%)	Δ Acc.	ρ_{TCD}	RMSE
None (nuGuidance)	70.3	—	8.7	0.095
Top-attended suppression	61.8	−8.5	2.6	0.139
Equal-mass random suppression	68.9	−1.4	7.9	0.101
Uniform attention	60.9	−9.4	1.2	0.151
Within-modality shuffle	65.1	−5.2	4.8	0.118

The controlled comparison

Top-mass vs. random suppression remove *equal attention mass*: −8.5 vs. −1.4 points. The pattern is functional, not descriptive.

Source: MotionScorer draft, Table 5, §5.3 (causal readout interventions).

The criterion tracks the damage

ρ_{TCD} falls monotonically with accuracy (8.7 → 7.9 → 4.8 → 2.6 → 1.2) — the strongest single piece of evidence for the filtering view.

13. Evidence III: The Frozen-Interface Efficiency Frontier

What does *not* touching the weights cost, and what does it buy (Table 8)?

Method	PairAcc. (%)	Backbone	Params	GPU-h	Ability Δ (avg/worst)
Direct prompting	54.2	frozen	0	0	0.00 / 0.00
Plain query readout	66.8	frozen	0.04M	21	0.00 / 0.00
nuGuidance (ours)	70.3	frozen	0.04M	23	0.00 / 0.00
LoRA scorer	71.5	updated	40.4M	118	-1.7 / -4.3
Full SFT scorer	72.4	updated	8.29B	760	-5.8 / -11.6

The correct claim (not “SOTA”)

Within 1.2 points of same-backbone LoRA at 1/1000 of the trainable parameters, 1/5 of the compute, and zero degradation of the backbone’s original abilities — where LoRA loses up to 4.3 points and SFT up to 11.6.

Caution: the draft’s Table 8 attaches $\pm$ intervals to deterministic columns (params, GPU-hours); those entries need fixing before submission and are omitted here.

Source: MotionScorer draft, Table 8, App. M (efficiency and capability retention).

14. Falsifiable Evaluation: Prediction, Kill Switches, Locked Test

Registered prediction (Eq. 13)

At matched teacher fit, query count, head capacity, budget, and checkpoint, adding ρ_{TCD} must

- ▶ raise held-out teacher-conditioned deflection, *and*
- ▶ improve held-out scoring, largest under distribution shift:

plain query readout < nuGuidance

Locked external test

WOD-E2E human ratings are never used for training, prompt selection, tuning, or model selection; no new human annotation anywhere.

The **[Exact]** / **[Assumption]** / **[Empirical]** tags used throughout this deck are the paper's own registry levels, applied slide by slide.

Source: MotionScorer draft, App. D.2 (Eq. 13), Table 7 (claim registry), §5.1 (locked WOD-E2E).

Kill switches (withdrawal conditions)

- ▶ Deflection rises *without* external held-out benefit $\Rightarrow$ prediction withdrawn.
- ▶ Comparable gains under *permuted* pseudo-scores $\Rightarrow$ prediction withdrawn; filtering demoted to “interpretive lens.”
- ▶ Probe ceiling fails to clear rule score and prompting $\Rightarrow$ observations (i)–(ii) narrowed.
- ▶ Matched-fit readouts fail to diverge $\Rightarrow$ observation (iii) dies.

Claim registry discipline (Table 7)

Every claim carries a level (Exact / Model / Empirical) and a required-evidence column; “levels are never upgraded in prose.”

15. Honest Ledger: What Each Step Actually Earns

The derivation is real, but its rungs are not equal. Colour-coded by provenance.

Step	Status
Non-interference / bit-identical context	[Exact] (architecture)
Readout $= \sum_i \alpha_i v_i$ (Eq. 2)	[Exact] (architecture)
Aggregation $\equiv$ tap-weighted filter	[Exact] (form identity)
Deflection optimum $\alpha^* \propto \Sigma^{-1}\theta$	[Exact] (classical, imported)
$\mathcal{A}_\phi$ realizability (Prop. 1)	[Assumption][$\rightarrow$] near-tautology
Signal+nuisance model (Eq. 7)	[Assumption] (sole declared premise)
ρ_{TCD} improves held-out quality	[Empirical] (Eq. 13; supported by slides 11–12)
Attention pattern causes the score	[Empirical] (Table 5 intervention, slide 12)

One-line verdict

Narrative architecture: *same tier* as LatentGauge. Mathematical depth: *one tier lower* (imports a closed form; names but does not characterize the space; the pivot is an assumption). Empirical discipline and honesty: *one tier higher*.

16. Where Deeper Derivation Would Pay — Three Executable Upgrades

A theorem is worth adding only if it *emits a curve an experiment can draw*. Three candidates, cheapest first.

A. Compensation lemma	B. Diagonal loading	C. Characterize $\mathcal{A}_\phi$
At zero training loss the readout is unidentifiable under head-absorbable maps; two fit-equivalent readouts diverge OOD by at least their attention-pattern distance. <i>Panel: Table 3 (equal RMSE, split OOD).</i>	Treat ε_D as robust MVDR loading; import Reed–Mallett–Brennan finite-sample loss. <i>Panel: deflection vs. batch size and vs. ε_D (non-monotone) — App. F is the empty slot reserved for it.</i>	Linearize E : softmax Jacobian $d\alpha = (\text{diag } \alpha - \alpha\alpha^\top)dk$ gives a reachable cone; predicts which Table 5 interventions hurt. <i>Highest novelty, hardest, local-only.</i>

Meta-advice

Do *one* to the bottom, not three halfway. LatentGauge’s depth comes from a single chain pushed all the way, not from theorem count. Priority: $A \rightarrow B \rightarrow C$.

17. Claim Status — Proven / Preliminary / Open

Proven now	Preliminary	Open
Trailing query is non-interfering (bit-identical).	ρ_{TCD} correlates with held-out quality (registered).	Does deflection <i>cause</i> OOD stability, or only correlate?
Readout = $\sum_i \alpha_i v_i$; form-identical to a tap filter.	Causal intervention: suppressing top-mass taps drops the score.	Structural characterization of $\mathcal{A}_\Phi$ beyond “image of E ”.
Classical $\alpha^* \propto \Sigma^{-1}\theta$ (imported).	Frozen backbone matches weight-modifying baselines at zero capability loss.	Compensation lemma: fit underdetermines the readout (currently empirical).

Reading

The left column is small but genuinely exact; the middle column is backed by filled main tables and a kill-switch registry; the right column is where a LatentGauge-grade theorem would move an entry leftward.

Source: MotionScorer draft, claim registry (Table 7) and §5 results.

References

- ▶ MotionScorer / nuGuidance. *Reading Trajectory Quality out of a Frozen Driving VLM*. ICLR 2027 draft (under review).
- ▶ J. Capon. *High-Resolution Frequency-Wavenumber Spectrum Analysis*. Proc. IEEE, 1969.
- ▶ H. L. Van Trees. *Optimum Array Processing (Detection, Estimation, and Modulation Theory, Part IV)*. Wiley, 2002.
- ▶ I. S. Reed, J. D. Mallett, L. E. Brennan. *Rapid Convergence Rate in Adaptive Arrays*. IEEE T-AES, 1974.
- ▶ Lou et al. *Trailing-Query Readout (SLQ) for Frozen Transformers*. 2026.
- ▶ Qwen2.5-VL Technical Report, 2025 (frozen driving-VLM backbone).

Companion analysis

Structural comparison to *LatentGauge* (measure-preserving planning-geometry attack): same narrative method (observations $\rightarrow$ formula similarity $\rightarrow$ optimization space $\rightarrow$ scenario-forced design); the two differ in whether the pivot step is a proof or a declared assumption.